

LLM4SDM

# Can smaller open-source LLMs measure shared decision making?

Observer OPTION12 coding in Dutch melanoma consultations

Tamara Wit\*, Lifeng Han\*, Carly Heipon, David Lindevelt,  
Anne Stiggelbout, Suzan Verberne

Leiden University Medical Center, Leiden University  
· 4D PICTURE consortium

\* co-first

arXiv:2607.06127 · development-phase pilot study



# The task is not to simply 'understand medicine'

## A clinical consultation

DR: We have several treatment options...

PT: What would each option mean for me?

DR: Let me check that this is clear...

### OPTION12 (refined)

12  
observer-rated  
behaviors  
Scores 0–4

### Human coding

Trained  
observers +  
consensus  
meetings

### Cost

Time-intensive  
and difficult to  
scale

**Can a privacy-preserving, locally deployable  
smaller LLM support this observer-coding  
workflow?**

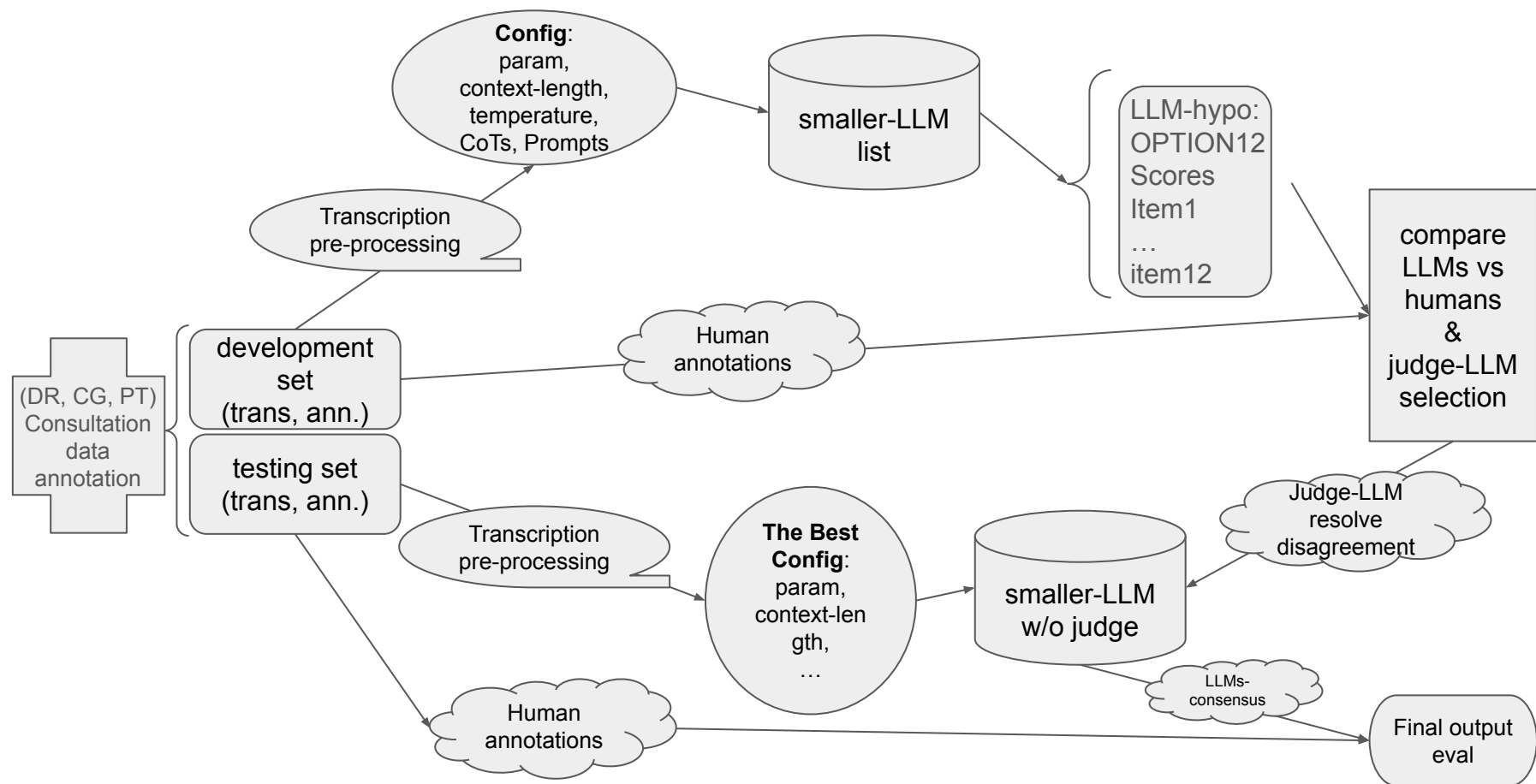
**NLP challenge: discourse structure · temporality ·  
speaker roles · communicative intent · evidence  
grounding**

# Option12 example

## 1. The clinician draws attention to an identified problem as one that requires a decision making process

This item assesses whether the clinician clearly indicates that a decision needs to be made about an identified problem. At the beginning of the consultation. Clear that the decision is shared. Shared is about the decision not specifically on the consultation.

| Score | Description                                                                                                                                                                                               | Example sentence                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | No attempt to indicate that a decision needs to be made.                                                                                                                                                  | No mentioning of any decision.                                                                                                                                                                                                                                                                                                                                                                                              |
| 1     | Rushed attempt, only briefly mentioned, vague/implicit reference to a decision.                                                                                                                           | <p>"So we'll go ahead with the next step."</p> <p>"We'll see what we can do about it."</p>                                                                                                                                                                                                                                                                                                                                  |
| 2     | <u>Acknowledges</u> that there is a decision <u>o</u> be made<br>Moderate effort to identify the decision point.                                                                                          | <p>"You and I will need to make a decision together about how to proceed."</p> <p>"A decision needs to be made today."</p>                                                                                                                                                                                                                                                                                                  |
| 3     | <u>Emphasizes the decision</u> and describes the shared process. <u>Clear</u> active attempt to focus <u>on decision</u> , <u>involves</u> the patient. Keywords such as: You and I, We, Together, Shared | <p>"This is an important decision, and I'd like us to take some time to go through it together, there are different paths we can take from here."</p> <p>So what we're going to do today is calmly go through what we've seen, what we've found, and what our possible proposal could be. We'll also discuss the fact that you have a choice in this, so we'll talk it through together, this is an important decision.</p> |
| 4     | (score of 3+)Describes the shared process and highlights intent to support the patient. Strong emphasis on need for decision and collaboration                                                            | "We are going to make a decision about your treatment today and I'm here to support you in making the choice that fits you best."                                                                                                                                                                                                                                                                                           |



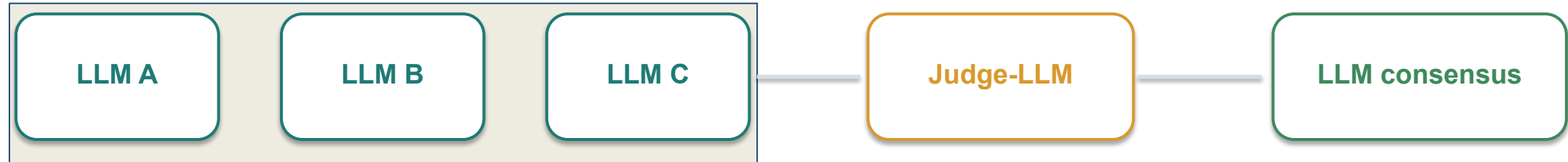
**LLM4SDM Overall Framework: framework: upper layer for development phase and lower layer for model testing**

# LLM4SDM mirrors a familiar annotation workflow

## HUMAN WORKFLOW



## LLM WORKFLOW



**Today: development-phase comparison + error analysis. Next: held-out testing + Judge-LLM consensus.**

# Development phase: 7 consultations, 5 models



## General-domain smaller-LLMs

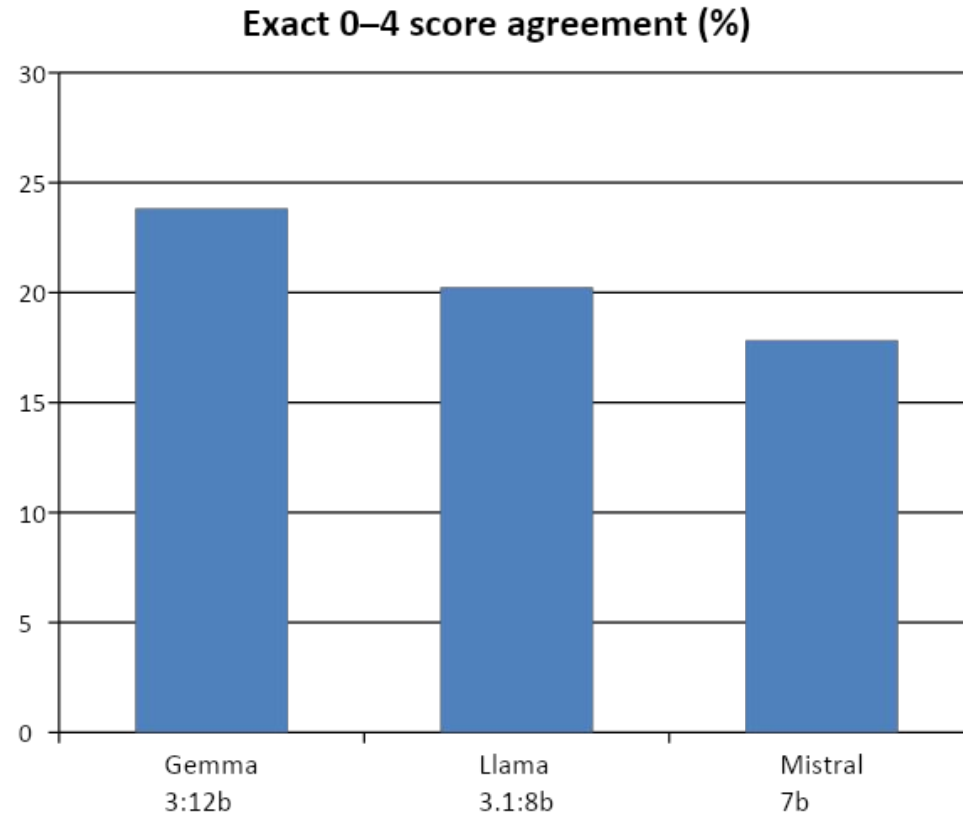
Gemma3:12b  
Llama3.1:8b  
Mistral7b

## Medical-domain smaller-LLMs

MedLlama2:7b  
Meditron7b

**Reference standard: two human coders → discussion → consensus OPTION12 scores**

# Gemma tracks human scoring trends best — but exact scores remain difficult



## Correlation with human annotations

Pearson  $r$

Gemma3:12b

**0.513**

Spearman  $\rho$

Gemma3:12b

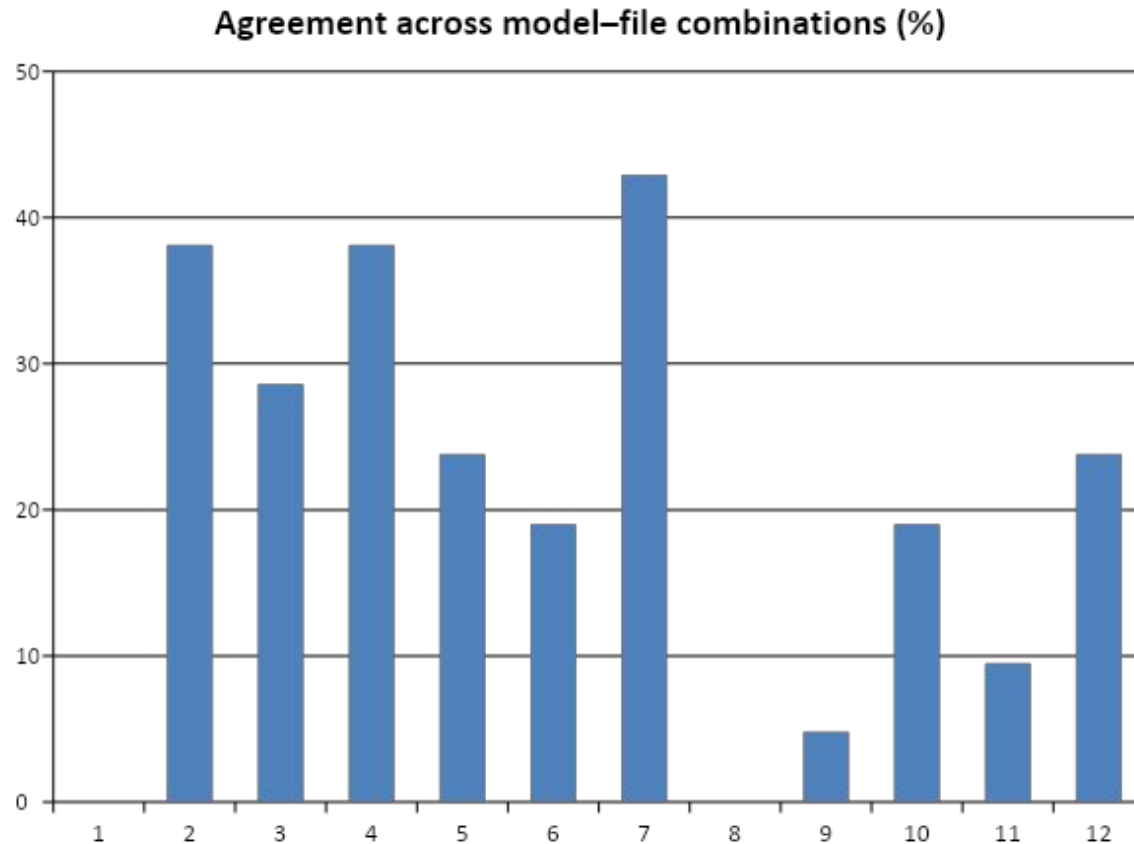
**0.587**

**Moderate correlation  $\neq$  reliable exact coding.**

**The models can follow broad trends while missing the precise score.**

**Current verdict: assistant, not autonomous annotator.**

# The hardest items expose discourse-level reasoning gaps



## Item 1 · 0%

“The clinician draws attention to an identified problem as one that requires a decision making process.”

**Needs temporal framing + discourse-structure reasoning**

## Item 8 · 0%

“The clinician checks that the patient has understood the information.”

**Needs interactional reasoning + verification recognition**

## Item 7 · 42.9%

Highest aggregate agreement in this pilot.

?: calculated on the exact match ratio with human coders

# A correct-looking score can still be wrong for the right reasons

## What the model must keep aligned

### SCORE

0–4 OPTION12 rating

### EVIDENCE

grounded transcript excerpt

### JUSTIFICATION

why evidence supports the item

## Observed failure modes

### Temporality

wrong stage of consultation

### Item interpretation

behavior definition misread

### Role confusion

speaker attribution swapped

### Score–evidence mismatch

score and rationale diverge

### Missing evidence

unsupported decision

### Hallucination

content not in transcript

**For clinical coding, explainability is part of correctness: score, evidence and justification must agree.**

# The medical models were not the medical solution

## General-domain models

Gemma3:12b  
Llama3.1:8b  
Mistral7b

Produced usable outputs for quantitative comparison.

## Medical-domain models

MedLlama2:7b  
Meditron7b

Frequent hallucination and instruction-following failures in our tested setup.

**Hypothesis: OPTION12 depends less on medical factual knowledge and more on discourse analysis, conversational reasoning and interaction assessment.**

# Three take-home messages

## 1 · Signal, not replacement

Smaller open-source LLMs capture some human scoring structure, but exact agreement is too low for autonomous OPTION12 coding.

## 2 · Errors are linguistic

The hard cases involve temporality, discourse structure, speaker roles, communicative intent and evidence grounding.

## 3 · Human-in-the-loop

The credible near-term role is annotation support and quality control, with local deployment supporting privacy.

## NEXT PHASE

Held-out  
15 transcripts

Prompt / model  
adaptation

Multi-model  
candidates

Judge-LLM  
consensus

Compare with  
human consensus

Thank you · Questions? · [arXiv:2607.06127](https://arxiv.org/abs/2607.06127)

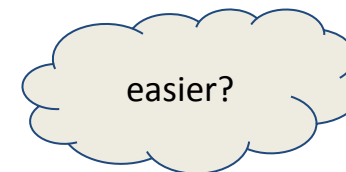


Paper  
arXiv:  
2607.06127

# Option12 example

## 8. The clinician checks that the patient has understood the information

This item assesses whether the clinician actively checks the patient's understanding of the information that has been shared. Checking should be after sharing information and before making a decision.

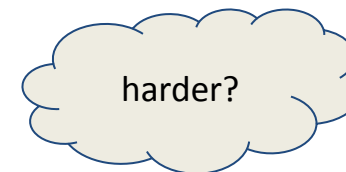


| Score | Description                                                                                                                                            | Example sentence                                                                                                                                                                                  |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | Does not check for understanding.                                                                                                                      | No attempt to check understanding.                                                                                                                                                                |
| 1     | Rushed/superficial attempt to check understanding. Closed questions that do not actively invite the patient to confirm or clarify their understanding. | "Okay? Is that clear?"and <u>Some</u> behave very calmly, others are more aggressive. The most important thing is that we take things step by step, but that you understand why we're doing them. |
| 2     | Explicitly asks if the patient understood the information.                                                                                             | "Does that make sense to you? Do you feel you've understood what we have discussed?"                                                                                                              |
| 3     | (Score of 2+) Checks with <u>patient</u> for understanding of information.                                                                             | "I'd like to check that you have understood the information about the possible options. Would you like to let me know what you understand about this issue?"                                      |
| 4     | (Score of 3+) <u>Asks patient</u> to summarize the provided information (teach-back method).                                                           | "Can you tell me in your own words what I've just explained?"                                                                                                                                     |

# Option12 example

## 12. The clinician indicates the need to review/defer the decision

This item assesses whether the clinician acknowledges that decisions may need to be revisited and provides an opportunity for the patient to review or revise the decision at a later time.



| Score | Description                                                                                       | Example sentence                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | No review or does not mention deferment .                                                         | No mention of review or reconsideration.                                                                                                                                   |
| 1     | Rushed attempt to review/defer the decision.                                                      | "You'll have/need a follow-up appointment."                                                                                                                                |
| 2     | Indicating the need for the patient to review/defer the decision.                                 | <p>"We can revisit this decision later if needed."</p> <p>"We will meet again in 3 weeks, and if you've changed your mind about treatment, we can talk about it"</p>       |
| 3     | Indicating the need to review/defer in a thoughtful and clear manner and explores patient doubts. | "If you have doubts later, we can talk again and reconsider. What concerns do you have right now?"                                                                         |
| 4     | Discusses patient doubts and asks <u>about need for reflection</u> time or deferment.             | "This is a big decision. It's okay to take time. Even after we start, we can always pause and reconsider. I will check with you to see how you're feeling about the plan." |

```
{ 'item_number': 10,  
  'item_description': 'Clinical assessment',  
  'score': 3,  
  'evidence': 'The patient was assessed by a dermatologist '  
              'and the diagnosis of melanoma was confirmed. '  
              'The Breslow thickness was 4 mm.',  
  'justification': 'This patient was seen in our clinic for '  
                  'the first time with a suspicious skin '  
                  'lesion on his left arm. A biopsy was '  
                  'performed, and the diagnosis of '  
                  'melanoma was confirmed by '  
                  'histopathology. The Breslow thickness '  
                  'was 4 mm.' },
```

Figure 2: Meditron Model Hallucination Example - repeated evidence and justification across predicted OPTIOIN12 items using the same score 3.

```

{'item_number': 12,
  'item_description': 'The patient has a history of '
                      'exposure to environmental toxins',
  'score': 1,
  'evidence': 'Environmental toxin exposure diagnosis',
  'justification': 'A history of exposure to environmental '
                  'toxins may increase the patient's risk '
                  'for developing cancer.'}],
'summary': {'mean_score': 10.0,
  'overall_quality': 'Good',
  'strengths': ['History of cancer diagnosis'],
  'weaknesses': ['Moedervlek on his back',
                 'Family history of cancer',
                 'History of smoking',
                 'Excessive alcohol consumption',
                 'Obesity',
                 'Lack of physical activity',
                 'Poor diet',
                 'Chronic stress',
                 'Poor sleep quality',
                 'Environmental toxin exposure'],
  'additional_comments': None}},

```

Figure 4: MedLlama Model Hallucination Example - giving score 1 for all OPTION12 items but mean score 10.

| OPTION12 item / summary                              | Gemma3:12b (%) | Llama3.1:8b (%) | Mistral7b (%) |
|------------------------------------------------------|----------------|-----------------|---------------|
| 1                                                    | <b>0</b>       | <b>0</b>        | <b>0</b>      |
| 2                                                    | <b>42.9</b>    | <b>42.9</b>     | 28.6          |
| 3                                                    | <b>42.9</b>    | 28.6            | 14.3          |
| 4                                                    | 28.6           | <b>57.1</b>     | 28.6          |
| 5                                                    | 14.3           | <b>28.6</b>     | <b>28.6</b>   |
| 6                                                    | <b>42.9</b>    | 14.3            | 0             |
| 7                                                    | 57.1           | 0               | <b>71.4</b>   |
| 8                                                    | <b>0</b>       | <b>0</b>        | <b>0</b>      |
| 9                                                    | <b>14.3</b>    | 0               | 0             |
| 10                                                   | <b>42.9</b>    | 0               | 14.3          |
| 11                                                   | 0              | <b>14.3</b>     | <b>14.3</b>   |
| 12                                                   | 0              | <b>57.1</b>     | 14.3          |
| Total consensus count<br>(over all items in 7 files) | <b>20</b>      | 17              | 15            |
| Mean consensus per file<br>(out of 12)               | <b>2.86</b>    | 2.43            | 2.14          |
| Mean consensus per file<br>(normalized, %)           | <b>23.83</b>   | 20.25           | 17.83         |

Table 2: Model-wise agreement per OPTION12 item between LLMs and human annotators across 7 files, with overall summary statistics. Bold indicates the best-performing model(s) for each row.

| <b>OPTION12 Item</b> | <b>Total Correct Count</b> | <b>Agreement (%)</b> |
|----------------------|----------------------------|----------------------|
| 1                    | 0                          | 0.0                  |
| 2                    | 8                          | 38.1                 |
| 3                    | 6                          | 28.6                 |
| 4                    | 8                          | 38.1                 |
| 5                    | 5                          | 23.8                 |
| 6                    | 4                          | 19.0                 |
| 7                    | 9                          | 42.9                 |
| 8                    | 0                          | 0.0                  |
| 9                    | 1                          | 4.8                  |
| 10                   | 4                          | 19.0                 |
| 11                   | 2                          | 9.5                  |
| 12                   | 5                          | 23.8                 |

Table 3: Total agreement counts and normalized agreement rates per OPTION12 item across all models (3) and files (7), i.e., out of 21 possible agreements per item.

Table 5: Qualitative analysis of LLM performance in shared decision making.

| Issue Type                                     | Item                                                                                                          | LLM Interpretation                                                                                                                                                                                                                                                                 | Human Annotation            | Consensus |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------|
| Temporality-management                         | Item 1: The clinician draws attention to an identified problem as one that requires a decision making process | LLM identifies the decision-making process at the end of the consultation rather than at the beginning. For example, Gemma and Mistral use the clinician’s statement about discussing treatment as evidence, even though the decision framing occurs too late in the consultation. | Human-1: 1; 0<br>Human-2: 2 |           |
| Misinterpreting the item/option                | Item 2: The clinician states that there is more than one way to deal with the identified problem              | Gemma cites explanation of melanoma and Breslow thickness as if it were discussion of multiple management options, but the quoted evidence does not present alternatives.                                                                                                          | Human-1: 1; 1<br>Human-2: 1 |           |
| Misinterpreting the item/option<br>(continued) | Item 1                                                                                                        | Llama 3.1 treats the clinician’s check of what was previously explained about melanoma as if it were introducing a decision process, while it is actually background clarification.                                                                                                | Human-1: 1; 0<br>Human-2: 2 |           |

Continued on next page

| Issue Type                                                  | Item                                                                                                               | LLM Interpretation                                                                                                                                                                        | Human Annotation                                              | Consensus |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-----------|
| Scoring right but justification/evidence not accurate       | Item 3: The clinician assesses the patient’s preferred approach to receiving information to assist decision making | Mistral assigns a low score appropriately, but the justification and evidence focus on explanation of Breslow thickness rather than the patient’s preferred way of receiving information. | Human-1: 1; Human-2: 1                                        |           |
| Scoring right but evidence not accurate                     | Item 4: Clinician lists all options, including doing nothing                                                       | Mistral’s justification correctly notes that treatment options were not explicitly discussed, but the selected evidence sentence is unrelated and appears arbitrary.                      | Human annotation: no listing; not detailed about wait-and-see | 1         |
| Justification / evidence right but score differs from human | Item 1 (File 130003)                                                                                               | Gemma, Llama, and Mistral correctly identify that the clinician introduces the treatment discussion, but they over-score it as full decision framing or shared decision making.           | Human: 1, 1, 1                                                | 1         |
| Missing justification/evidence                              | Item 3 (File 130003)                                                                                               | Gemma assigns score 1 and states that there is no explicit assessment of the patient’s preferred learning style, but provides no actual evidence excerpt.                                 | Human: 1, 1, 1                                                | 1         |

Continued on next page

| Issue Type                                               | Item                                                                         | LLM Interpretation                                                                                                                                                                             | Human Annotation | Consensus |
|----------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------|
| Role misunderstanding                                    | Item 8: The clinician checks that the patient has understood the information | Mistral uses a caregiver utterance as if it were spoken by the clinician, leading to an incorrect attribution of understanding-check behavior.                                                 | Human: 0, 0, 0   | 0         |
| Hallucinated evidence                                    | General issue                                                                | In some cases, the LLM produces evidence or statements that are not grounded in the transcript at all.                                                                                         | –                | –         |
| Evidence / justification partly right but score too high | Item 1 / later example from File 130003                                      | The clinician asks for agreement with surgery, and the evidence is relevant, but the model over-interprets this as balanced shared decision making, whereas the interaction is more directive. | Human: 2, 2, 2   | 2         |
| Evidence / justification okay but score too high         | Continued example                                                            | Llama scores the clinician’s brief acknowledgment of patient anxiety too highly, although the concern is only superficially explored.                                                          | Human: 1, 1, 1   | 1         |